

ARENA OF RATINGS: REAL-TIME MATCHMAKING ENGINE

Name: Syed Soban Ahmed Shah

Roll No: 37

Group: "B"

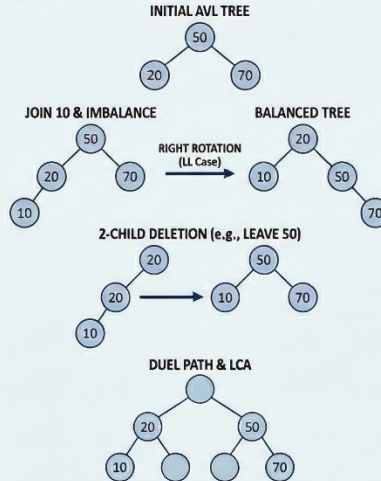
PROBLEM STATEMENT

- Build a high-performance matchmaking engine for a competitive gaming environment.
- Manage unique integer ratings for a dynamic player base.
- Handle up to 200,000 commands efficiently with real-time query response.
- Provide player info, sorted range listing, and structural diagnostics.

PROPOSED SOLUTION

- **Custom AVL Tree:** Implemented a self-balancing binary search tree for $O(\log N)$ efficiency.
- **Augmented Structure:** Nodes store metadata (height, subtree size) for RANK and KTH queries.
- **Strict Ordering:** Engine maintains a strict order based exclusively on unique player ratings.

GRAPHICAL VIEW OF ENGINE OPERATIONS



WORKING METHODOLOGY

- **Structure Setup:** Pointer-based AVL tree for player records (name, rating, HP).
- **Dynamic Updates:** Supports real-time JOIN/LEAVE commands with rebalancing.
- **Deletion Logic:** Correctly handles 0, 1, and 2-child cases; maintains order.
- **Matchmaking Logic:** Implements MATCH command, ties default to lower rating.
- **Advanced Analytics:** Provides path distances via DUEL command using LCA.

CONCLUSION

- Successfully processed 200,000 commands using a custom-built AVL tree.
- Achieves high efficiency and reliability for competitive matchmaking, ranking, and data management.
- Implementation meets all strict structural and formatting requirements.